Evaluation model for Real Estate Investment Environment Based on SVM

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Abstract

The environment is the key to the success of real estate investment. Therefore, it is very important for the investment environment evaluation of real estate. Firstly, the basic theory of support vector machine for regression (SVR) was introduced, and then a SVM to evaluate the investment environment of real estate was built. Through simulating and comparing with the method of BP neural network, the method is proved effectiveness.

1. Introduction

With the process of Chinese rapid development in the real estate industry, there have been some phenomena, such as “hot zone” or “hot land”, “housing with a high vacancy rate”, the imbalance in real estate development. These problems are connected with underestimation and the wrong choice of the real estate investment environment to certain extent. So doing the correct evaluation to the investment environment is very important.

The investment environment of real estate is a external condition that produces the favorable or unfavorable impact on achieving the investment objective in a certain region of investment(national, regional, city or neighborhood)and a certain period of time [1].At present, the common methods to evaluate the investment environment include: (1) Primary factor scoring method, including Multi-factor rating scale method, Psychological assessment method and hot and cold factor analysis method ,etc.(2) Multi-factor system assessment method, including multi-factor assessment methods, comprehensive evaluation methods, etc.(3)Comprehensive quantitative methods, including AHP level analysis method, fuzzy comprehensive evaluation method, BP neural network fuzzy comprehensive evaluation method, gray fuzzy comprehensive evaluation method, grey-clustering analysis and so on. These methods are explored based on analysis of domestic and foreign investment environment needs, whose basic principles have a wider adaptability [2].

Statistical Learning Theory (SLT) is a kind of statistical theory which proposed in the late-1970s in view of the limited sample by V.N.Vapnik. Support Vector Machine (SVM), is the most successful realization of this theory so far [3, 4]. The SVM bases on the structure risk minimum principle of the statistics theory, and simultaneously considerers the training error and the pan-ability. Seek best result between the model complexity and learning capability according to the limited sample information. It displays much unique superiority in solving pattern recognition problems on the little sample, misalignment, high dimension, partial minimum. The SVM theory already successfully deals with the return issue (time series analysis) and the pattern recognition (classified question, distinguishable analysis) and so on at present. Support Vector Regression, called SVR, which may approach the nonlinear function according to the random precision, and has the merit of overall minimum point and quick convergence rate, already successfully applies in the weather predict ,the subsoil water level predict , the load forecast of electrical power system as well as stock price forecast. This article, basing on SVM, proposes a new method of estates investment environmental evaluation, and confirms its feasibility by the example.

2. Index system of real estate investment environment evaluation

Investment environment evaluating system is the basis and the standard on the synthetic evaluation of the investment environment. Establishing it should conform to the principle of comprehensive system, concise scientific, stability and comparability, easy operation. Therefore, the article bases on the actual situation of investment profession. Using systems engineering theory, it establishes a set of investment environment appraisal system which has some merits,

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such as: complete structures, rich content, easy operation, and so on. The primary factor has the natural environment, the economic environment, the policy environment, the social culture environment. These four aspects also have the multistage child factors, which shown in Table 1.

Table 1 Index system of real estate investment environment evaluation

The investment environment of real estate	Natural environment	Infrastructure X_1
		Transportation X_2
		Environment status X_3
		Land statusX_4
		Location X_5
	Economic environment	supply and demand of real estate X_6
		Transmission service of real estate X_7
		Land price X_8
		People's income level X_9
		Economic growth level X_{10}
		Similar estate distribution X_{11}
	Policy environment	Real estate policy X_{12}
		Government working efficiency X_{13}
		Urban planning control X_{14}
		Tax X_{15}
	Social culture environment	Regional population quantity X_{16}
		Regional population quality X_{17}
		Family population constitute X_{18}
		Consume culture and idea X_{19}
		Social order factor X_{20}

3. Investment environment evaluation of real estate based on SVM

Based on SVM, the evaluation model of real estate investment environment was set up.

Let D be the sample collection, and

$$D=\left\{\left(x_i,y_i\right)|\text{i=1,2,…, n}\right\}x_i\in R^n, y_i\in R$$

Where x_i is the ith study sample's input value, and the index attribute value of the ith real estate investment environment evaluation, and it is the n dimension vector, $x_i=\left(x_i^1,x_i^2,...,x_i^n\right)$, y_i is corresponding to the target output value, and the corresponding value of real estate investment environment evaluation .

3.1 Linear regression problems

The linear regression equation:

$$f\left(x\right)=\left\langle\omega,x\right\rangle+b \tag{1}$$

Where $\omega\in R^n$ is the weight vector, $b\in R$ is the hold value.

First define the insensitive loss function is ε -insensitive, its form is:

$$L_\varepsilon\left(y\right)=\begin{cases}0,\left|f\left(x\right)-y\right|\le\varepsilon\\ \left|f\left(x\right)-y\right|+\varepsilon,\left|f\left(x\right)-y\right|>\varepsilon\end{cases} \tag{2}$$

That is, if the difference of the target y and the estimated value of the function $f\left(x\right)$ is less than ε, the loss is 0.

SVR problem can be translated into solving mathematical programming at the following (Introduced of slack variable ξ_i,ξ_i^*):

$$\min_{\omega,b,\xi_i,\xi_i^*}\left\|\omega\right\|^2/2+C\sum_{i=1}^n\left(\xi_i+\xi_i^*\right) \tag{3}$$

$$s.t.\quad \begin{aligned}&\left(\left(\omega\cdot x_i\right)+b\right)-y_i\le\omega+\xi_i\\ &y_i-\left(\left(\omega\cdot x_i\right)+b\right)\le\omega+\xi_i^*\\ &\xi_i,\xi_i^*\ge0,(i=1,2,\cdots,n)\end{aligned} \tag{4}$$

Through Lagrange transform, to be its dual problem, so:

$$\max_{\alpha,\alpha^*}\left[-\frac{1}{2}\sum_{i=1}^n\sum_{j=1}^n\left(\alpha_i-\alpha_i^*\right)\left(\alpha_j-\alpha_j^*\right)\left\langle x_i,x_j\right\rangle+\sum_{i=1}^n y_i\left(\alpha_i-\alpha_i^*\right)-\varepsilon\sum_{i=1}^n\left(\alpha_i+\alpha_i^*\right)\right] \tag{5}$$

$$s.t.\quad \begin{aligned}&\sum_{i=1}^n\left(\alpha_i-\alpha_i^*\right)=0\\ &0\le\alpha_i,\alpha_i^*\le C,(i=1,2,\cdots,n)\end{aligned} \tag{6}$$

In this equation, C is the positive constant, called penalty parameter.

Solving the programming problem on the above, to be the Lagrange multiplier α_i,α_i^*, then, the coefficient in the regression equation:

$$\omega=\sum_{i=1}^n\left(\alpha_i-\alpha_i^*\right)x_i \tag{7}$$

Through the Karush-Kuhu-Tucker conditions, when $0<\alpha_i<C$, $\xi_i=0$, $\left(\left(\omega\cdot x_i\right)+b\right)-y_i\le\omega+\xi_i$ becomes the equation,

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so b can be obtained; When $0<\alpha_i^*<C$, $\xi_i^*=0$, $y_i-\left(\left(\omega\cdot x_i\right)+b\right)\le\omega+\xi_i^*$ becomes the equation, so b can be obtained, too.

3.2 Nonlinear regression problems

The article discusses the investment environment evaluation of real estate for nonlinear regression problems, through introducing the appropriate kernel function $K\left(x,y\right)$, the input sample space can become into another feature space by non-linear conversion, and in this space do the linear estimation by the function in the original space.

There are commonly used kernel functions:

Linear kernel function:

$$K\left(x,y\right)=xy \tag{8}$$

Polynomial kernel function:

$$K\left(x,y\right)=\left(xy+1\right)^d \quad d=1,2,\cdots \tag{9}$$

Radial Basis kernel function:

$$K(x,y)=\exp\left[-\frac{\left\|x-y\right\|^2}{2\sigma^2}\right] \tag{10}$$

Sigmoid inter production function:

$$K\left(x,y\right)=\tanh\left(v\left(xy\right)+c\right) \tag{11}$$

Where c is the kernel parameter.

Here, the Radial Basis kernel function is adopted.

The nonlinear regression equation:

$$f\left(x\right)=\sum_{i=1}^{n}\left(\alpha_i-\alpha_i^*\right)K\left(x_i,x\right)+b \tag{12}$$

Solving the following planning problems:

$$\max_{\alpha,\alpha^*}\left[-\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left(\alpha_i-\alpha_i^*\right)\left(\alpha_j-\alpha_j^*\right)K\left(x_i,x_j\right)+\sum_{i=1}^{n}y_i\left(\alpha_i-\alpha_i^*\right)-\varepsilon\sum_{i=1}^{n}\left(\alpha_i+\alpha_i^*\right)\right] \tag{13}$$

$$s.t. \quad \sum_{i=1}^{n}\left(\alpha_i-\alpha_i^*\right)=0 \tag{14}$$

$$0\le\alpha_i,\alpha_i^*\le C,(i=1,2,\cdots,n)$$

3.3 Model's appraisal principles and the step to realize

Based on the SVM, the estates investment environmental evaluation model takes the factors which affect the estates investment environment as the training samples which support vector machines. It constructs appraises the machine ideally by modifying model and the parameter trains of nuclear function.

The step is as follows:

Step 1: establish indicator system of estates investment environmental evaluation. The article uses indicator system like table 1.

Step 2: take the certain amount of project whose result has been made as the sample, and then, establish the training sample collection.

Step 3: make out model's parameter ε, choose the nuclear function parameter σ and penalty parameter C.

Step 4: input target attribute value, carry on the appraisal to the estates investment environment.

4. Case analyses

This article uses the data in reference [5] to carry on the training, as Table 2 shown. Takes the value of X_1 to the X_{20} in SVR indicator system as the input, and takes the score at the last row as the expectation output of SVR, then, carries on the simulation with SVR. The test result divides into 4 levels: 1-0.8 for outstanding, 0.8-0.7 for good, 0.7-0.5 for general, 0.5-0 for bad. Selects the first 12 achievement as training sample, and selects other 4 achievement as examination sample. Takes the insensitive loss function measure value $\varepsilon=0.0001$, selects penalty parameter $C=700$ by alternately conform, selects the radial direction base nuclear function parameter $\sigma=1.52$. Carries on the simulation by SVR under the Matlab6.5, and then compares with the BP neural network simulation result in reference [5] which is shown in Table 3.

Table 3 Training results

No.	Expect output	SVR		BP Neural network	
		Training results	Relative Error %	raining results	Relative Error %
1	0.861	0.8609	0.0116	0.8612	0.0232
2	0.604	0.6035	0.0828	0.6029	0.1821
3	0.34	0.3396	0.1176	0.3408	0.2353
4	0.488	0.4883	0.0615	0.4894	0.2869
5	0.713	0.7146	0.2244	0.7185	0.7714
6	0.931	0.9328	0.1933	0.9193	1.257
7	0.81	0.8107	0.0864	0.8094	0.0741
8	0.641	0.6421	0.1716	0.6343	1.0452
9	0.683	0.6843	0.1903	0.689	0.8785
10	0.827	0.829	0.2418	0.8289	0.2297
11	0.647	0.6472	0.0309	0.6511	0.6337
12	0.727	0.7219	0.7015	0.7196	1.0179

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From the above simulation result, we can know that the output result which is got by using simulation of examination sample is the same as expectation output.

The biggest relative error is 1.8913%, and the smallest relative error is 0.0652%. Comparing the neural network method, the total relative error, got by using simulation of examination sample, is small (see Table 4). It explains that SVM is more accurate and effective than the BP neural network on the appraisal result.

Table 4 Comparison of test results and classification

No.	Expect output	SVR		BP Neural network	
		Test results	Error	Test results	Error
13	0.46	0.4513	1.8913	0.4509	1.9783
14	0.817	0.8247	0.9425	0.8369	2.4357
15	0.766	0.7665	0.0652	0.7567	1.2141
16	0.63	0.6381	1.2857	0.6614	4.9841

Table 2 Learning samples

No.	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}	X_{13}	X_{14}	X_{15}	X_{16}	X_{17}	X_{18}	X_{19}	X_{20}	*scores*
1	0.7	1	1	0.7	0.7	1	0.7	0.7	1	1	0.7	1	0.7	0.7	0.7	0.7	1	1	1	1	0.861
2	0.5	0.7	0.5	0.7	0.5	0.5	0.5	0.7	0.5	0.7	0.7	0.5	0.7	0.5	0.7	0.7	0.7	0.7	0.7	0.5	0.604
3	0.5	0.5	0.3	0.5	0.3	0.3	0.3	0.5	0.3	0.5	0.3	0.3	0.3	0.5	0.3	0.3	0.3	0.3	0.3	0.1	0.34
4	0.5	0.5	0.5	0.5	0.7	0.3	0.3	0.5	0.5	0.7	0.7	0.5	0.3	0.5	0.7	0.7	0.7	0.7	0.3	0.5	0.488
5	0.5	0.5	0.5	0.7	0.7	0.5	1	0.7	1	0.7	0.7	1	1	0.7	0.7	0.7	1	0.7	0.7	0.1	0.713
6	1	0.7	1	1	1	1	1	0.7	1	1	0.7	1	1	0.7	0.7	0.7	1	0.3	0.7	1	0.931
7	1	0.7	1	1	1	1	1	0.7	1	1	0.7	1	1	0.7	0.5	0.7	1	1	0.7	0.1	0.81
8	0.7	0.7	0.5	0.7	0.7	0.5	0.7	0.5	0.7	0.7	0.5	0.5	0.7	0.7	0.7	0.7	0.7	0.7	1	0.5	0.641
9	0.7	0.7	1	0.7	0.7	0.5	0.7	0.5	0.7	0.7	0.7	0.5	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.5	0.683
10	0.7	0.7	1	1	0.7	0.5	1	0.7	1	1	0.7	1	1	0.7	0.7	0.7	1	0.7	1	0.5	0.827
11	0.7	0.5	0.5	0.7	0.7	0.5	0.7	0.7	0.5	0.7	0.7	0.5	0.7	0.7	0.7	0.5	0.7	0.7	0.7	1	0.647
12	0.7	0.7	1	0.7	0.7	0.5	1	0.7	1	0.7	0.7	1	1	0.7	0.7	0.7	1	0.7	0.7	0.5	0.727
13	0.7	0.7	0.3	0.5	0.3	0.5	0.3	0.5	0.5	0.5	0.7	0.5	0.3	0.7	0.7	0.5	0.7	0.3	0.3	0.1	0.46
14	0.7	1	1	1	0.7	1	1	0.7	1	1	0.7	1	0.7	0.7	0.7	0.7	0.7	0.7	1	0.5	0.817
15	0.7	0.7	0.5	1	0.7	1	1	0.7	1	1	0.7	1	1	0.7	0.7	0.7	0.7	0.7	0.7	0.5	0.766
16	0.7	0.7	0.5	0.7	0.7	0.5	0.7	0.5	0.5	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.3	1	0.1	0.63

5. Conclusion

SVM, proposed early in the classified question, is gradually promoted to the misalignment regression estimate, and has made the good progress. SVM, which is used in the estates investment environmental evaluation in this article, has avoided sample complex. SVM is a simple effective method which making the rightly appreciate on real estates investment environment.

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